

IPL PERFORMANCE ANALYSIS DASHBOARD (2008–2017)

Internship Project Report

**Submitted in partial fulfilment of the
Business Analytics Internship Program**

Project Title: IPL Data Analysis Using Microsoft Power BI

Tools Used:

- **Microsoft Power BI**
 - **Power Query**
 - **Microsoft Excel**

Dataset: IPL Matches & Deliveries Dataset (2008–2017)

Domain: Business Analytics

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Executive Summary

The **IPL Analysis Dashboard (2008–2017)** was developed using **Microsoft Power BI** to transform raw IPL match data into an interactive and easy-to-understand visual report. The project aims to analyse historical IPL data and identify meaningful trends related to team performance, player statistics, toss decisions, match outcomes, and venue performance. By presenting the data through dashboards, users can quickly interpret large volumes of information and make informed decisions based on key performance indicators.

The analysis covers **10 IPL seasons (2008–2017)** and includes **636 matches, 202 players, and 14 teams**. The dashboard is divided into three main sections: an **Overview Dashboard**, which provides tournament-level insights; a **Team Performance Dashboard**, which compares teams based on wins, tosses, venues, and seasonal performance; and a **Player Performance Dashboard**, which highlights the leading batsmen, bowlers, and fielders during the selected period.

The analysis indicates that the **Mumbai Indians** emerged as the most successful team based on total match victories. **Suresh Raina** scored the highest number of runs among the displayed players, while **Lasith Malinga** recorded the highest number of wickets. The dashboard also shows that teams chose to **field first in approximately 57% of matches**, suggesting a preference for chasing targets. Furthermore, the majority of IPL matches ended with normal results, with very few ties or matches ending without a result. Mumbai hosted the highest number of matches, making it one of the tournament's most significant venues.

Overall, the dashboard demonstrates how **Business Intelligence (BI)** tools can convert complex sports datasets into meaningful visual insights. The findings can support coaches, team management, analysts, tournament organizers, and cricket enthusiasts in understanding historical performance patterns and making data-driven strategic decision

Dashboard Overview

The **IPL Analysis Dashboard (2008–2017)** is an interactive Business Intelligence solution developed using **Microsoft Power BI**. It is designed to transform raw IPL match data into meaningful visual insights, enabling users to analyse tournament performance efficiently. The dashboard combines key performance indicators (KPIs), charts, and filters to provide a comprehensive overview of the Indian Premier League across ten seasons.

The dashboard is structured into three dedicated sections, each focusing on a different aspect of IPL performance:

- **Overview Dashboard:** Presents tournament-level statistics, including total matches, participating teams, players, toss decisions, match results, and host city analysis.
- **Team Performance Dashboard:** Analyses team-wise performance by comparing total wins, toss victories, venue performance, and season-wise participation.
- **Player Performance Dashboard:** Highlights the performance of leading batsmen, bowlers, and fielders through key statistical comparisons.

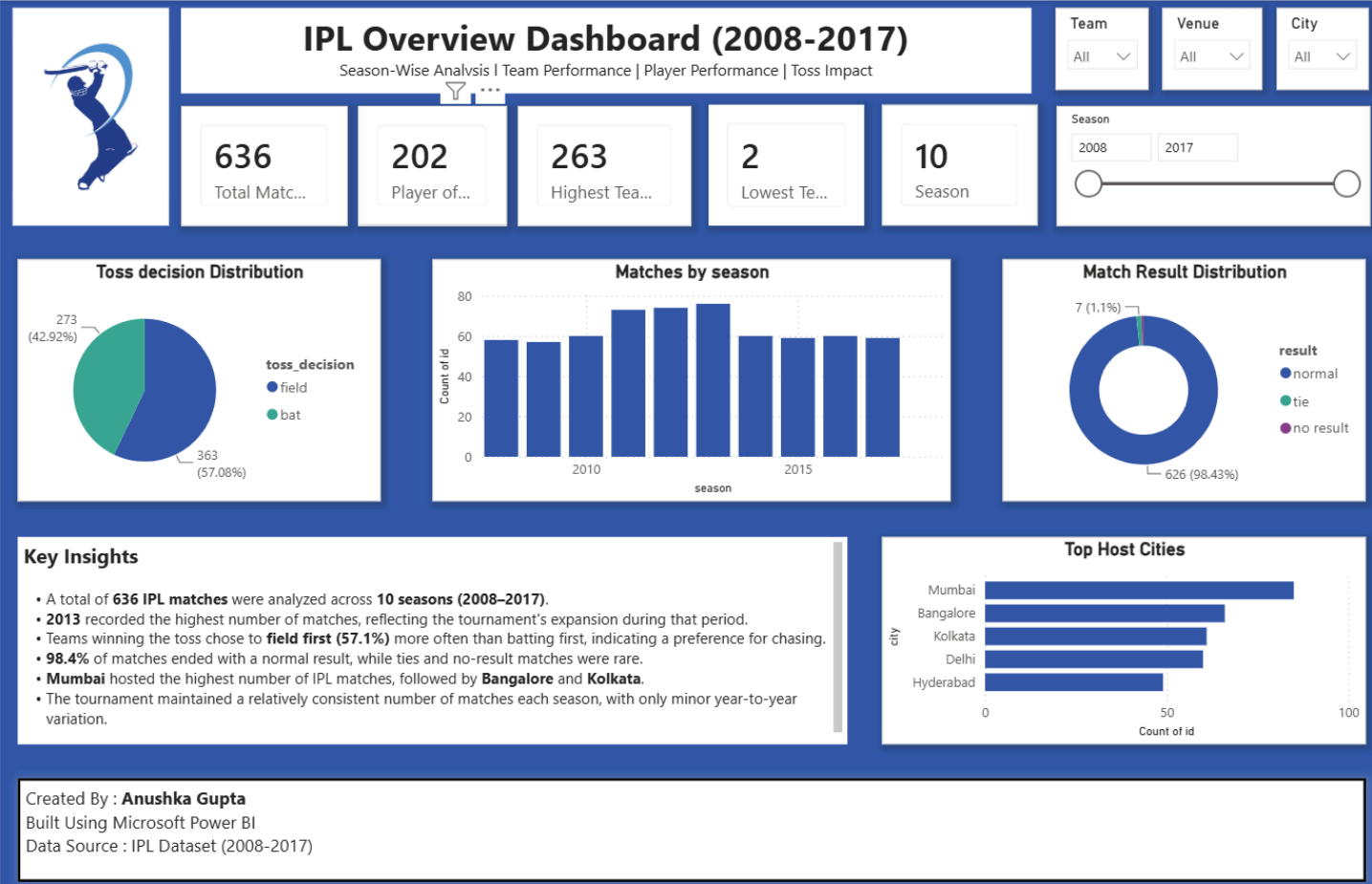
The dashboard also includes interactive filters for **Team**, **Season**, **Venue**, and **City**, allowing users to customize the analysis and compare different scenarios. These interactive features make the dashboard suitable for coaches, analysts, team management, and cricket enthusiasts seeking quick and meaningful insights.

Dataset Summary

Parameter	Details
Dataset	IPL Match Dataset
Seasons Covered	2008–2017
Total Seasons	10
Total Matches	636
Total Teams	14
Total Players	202
Visualization Tool	Microsoft Power BI

Key Features of the Dashboard

- Interactive KPI cards for quick performance monitoring.
- Dynamic filters (slicers) for customized analysis.
- Visualizations including bar charts, pie charts, and line charts.
- Dashboard navigation across Overview, Team Performance, and Player Performance.
- A dedicated **Key Insights** section on every dashboard page for quick interpretation of results.



IPL Overview Dashboard Analysis

The IPL Overview Dashboard provides a comprehensive summary of the tournament between **2008 and 2017**, presenting key statistics, seasonal trends, toss decisions, match outcomes, and venue distribution. The dashboard enables users to quickly understand the overall performance of the league before exploring detailed team and player analyses.

Key Performance Indicators (KPIs)

The dashboard highlights the following tournament statistics:

- **Total Matches:** 636
- **Total Seasons:** 10 (2008–2017)
- **Total Teams:** 14
- **Total Players:** 202
- **Highest Team Score:** 248 runs
- **Lowest Team Score:** 49 runs

These KPIs provide a quick snapshot of the tournament and establish the scope of the analysis.

Season-wise Match Distribution

The line chart indicates that the number of IPL matches increased steadily after the inaugural season, with the highest number of matches being played during **2013**. After this peak, the tournament maintained a relatively consistent schedule, reflecting the league's stability and growth over the years.

Toss Decision Analysis

The pie chart shows that:

- **57.1%** of teams chose to **field first** after winning the toss.
- **42.9%** of teams chose to **bat first**.

This suggests that most captains preferred chasing targets, likely because they believed it offered a tactical advantage under varying pitch and weather conditions.

Match Result Analysis

The dashboard reveals that:

- **98.4%** of matches ended with a **normal result**.
- Only a very small percentage of matches resulted in **ties** or **no results**.

This indicates that IPL matches were generally completed successfully, providing a reliable dataset for performance analysis.

Host City Analysis

Among all host cities, **Mumbai** hosted the highest number of IPL matches, followed by **Bangalore** and **Kolkata**. This reflects the importance of these cities as major cricketing venues and their consistent role in hosting IPL fixtures throughout the selected seasons.

Key Insights

- The IPL expanded steadily and reached its highest number of matches in **2013**.
- Teams showed a clear preference for **fielding first**, indicating the popularity of chasing strategies.
- Almost every match produced a definitive result, highlighting the consistency of the tournament.
- Mumbai emerged as the leading host city, emphasizing its significance in the IPL schedule.
- The tournament maintained a balanced level of participation across ten successful seasons.

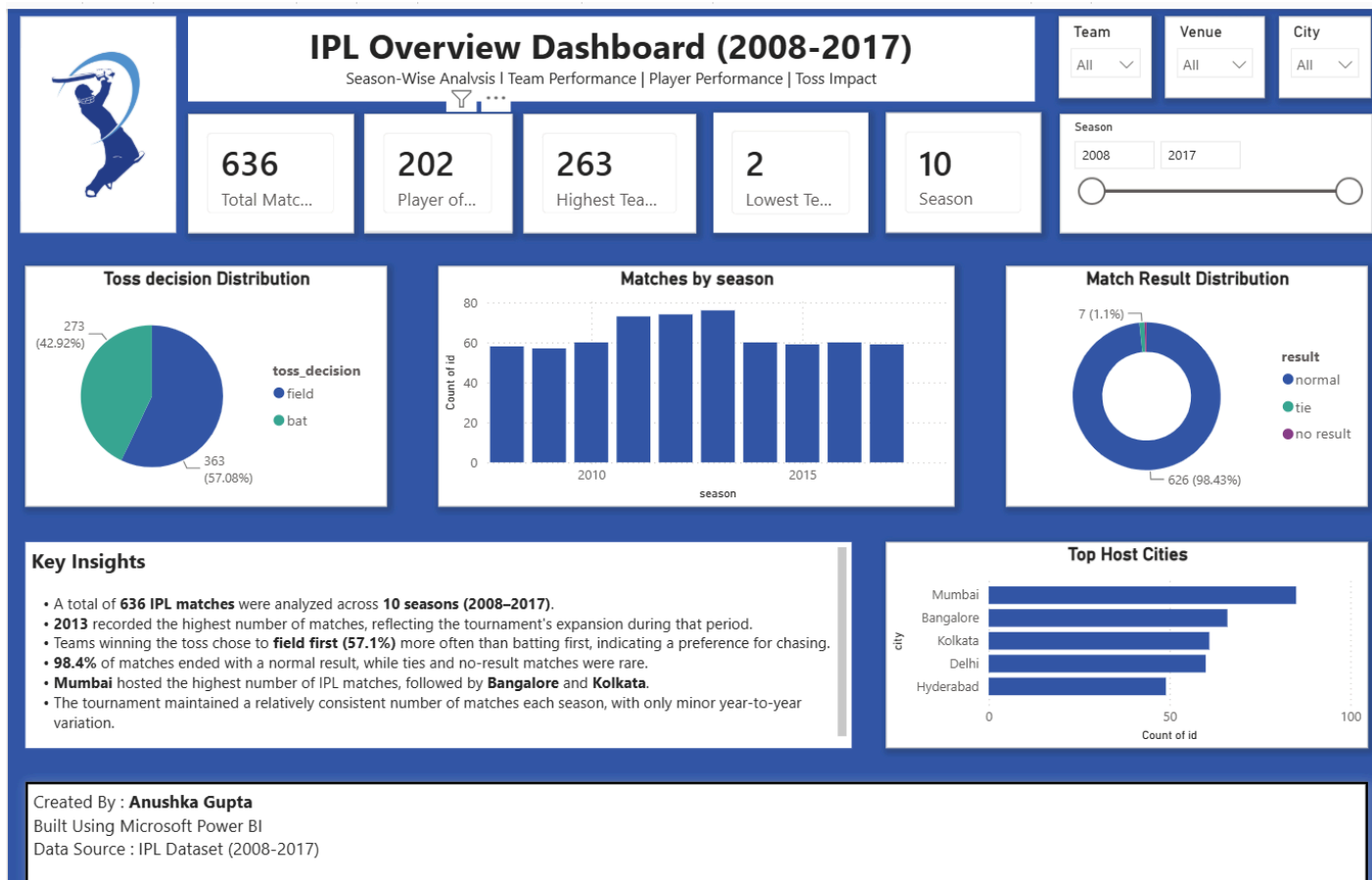


Figure 2: IPL Overview Dashboard (Analysis)

Team Performance Dashboard Analysis

The **Team Performance Dashboard** provides a detailed comparison of all IPL teams based on match victories, toss wins, venue performance, and season-wise participation. This analysis helps identify the most successful teams, understand performance trends, and evaluate the influence of home venues and toss outcomes on match results.

Overall, Team Performance

The dashboard shows that the **Mumbai Indians (MI)** recorded the **highest number of match victories** during the 2008–2017 period, demonstrating consistent performance across multiple seasons. **Chennai Super Kings (CSK)** and **Kolkata Knight Riders (KKR)** also emerged as strong performers, maintaining competitive records throughout the tournament.

In contrast, teams such as **Pune Warriors**, **Kochi Tuskers Kerala**, and **Gujarat Lions** registered comparatively fewer victories due to their shorter participation in the IPL.

Toss Winning Performance

The analysis indicates that **Mumbai Indians** won the highest number of tosses among all teams. However, winning the toss did not always guarantee victory, suggesting that team performance, player execution, and match conditions played a more significant role in determining the final outcome.

Venue-wise Performance

The dashboard highlights each team's most successful home venue. For example, **Mumbai Indians** achieved the greatest success at **Wankhede Stadium**, where they secured a large share of their victories. This demonstrates the advantage of familiar playing conditions and strong home support.

Season-wise Participation

The season-wise comparison shows that the number of matches increased steadily in the early years of the IPL and remained relatively consistent after 2013. This

consistency reflects the league's structured scheduling and continued growth during the selected period.

Key Insights

- **Mumbai Indians** were the most successful team in terms of total match wins.
- **Chennai Super Kings** and **Kolkata Knight Riders** consistently remained among the top-performing franchises.
- Winning the toss provided an advantage in some matches, but it was **not the sole factor** influencing match results.
- Teams generally performed better at their **home venues**, highlighting the importance of familiar conditions and crowd support.
- The IPL maintained a balanced competitive structure, with several teams recording strong performances across different seasons.

Recommendations

Based on the team performance analysis, the following recommendations can be made:

- Teams should use **historical performance data** to develop match strategies against specific opponents.
- Greater emphasis should be placed on **venue-specific planning**, as home conditions can significantly influence performance.
- Toss decisions should be combined with **pitch reports and weather conditions** rather than relying on a fixed strategy.
- Franchise management should focus on maintaining **consistent team combinations** to achieve sustained success over multiple seasons.
- Performance dashboards should be updated regularly to monitor team progress and support data-driven decision-making.

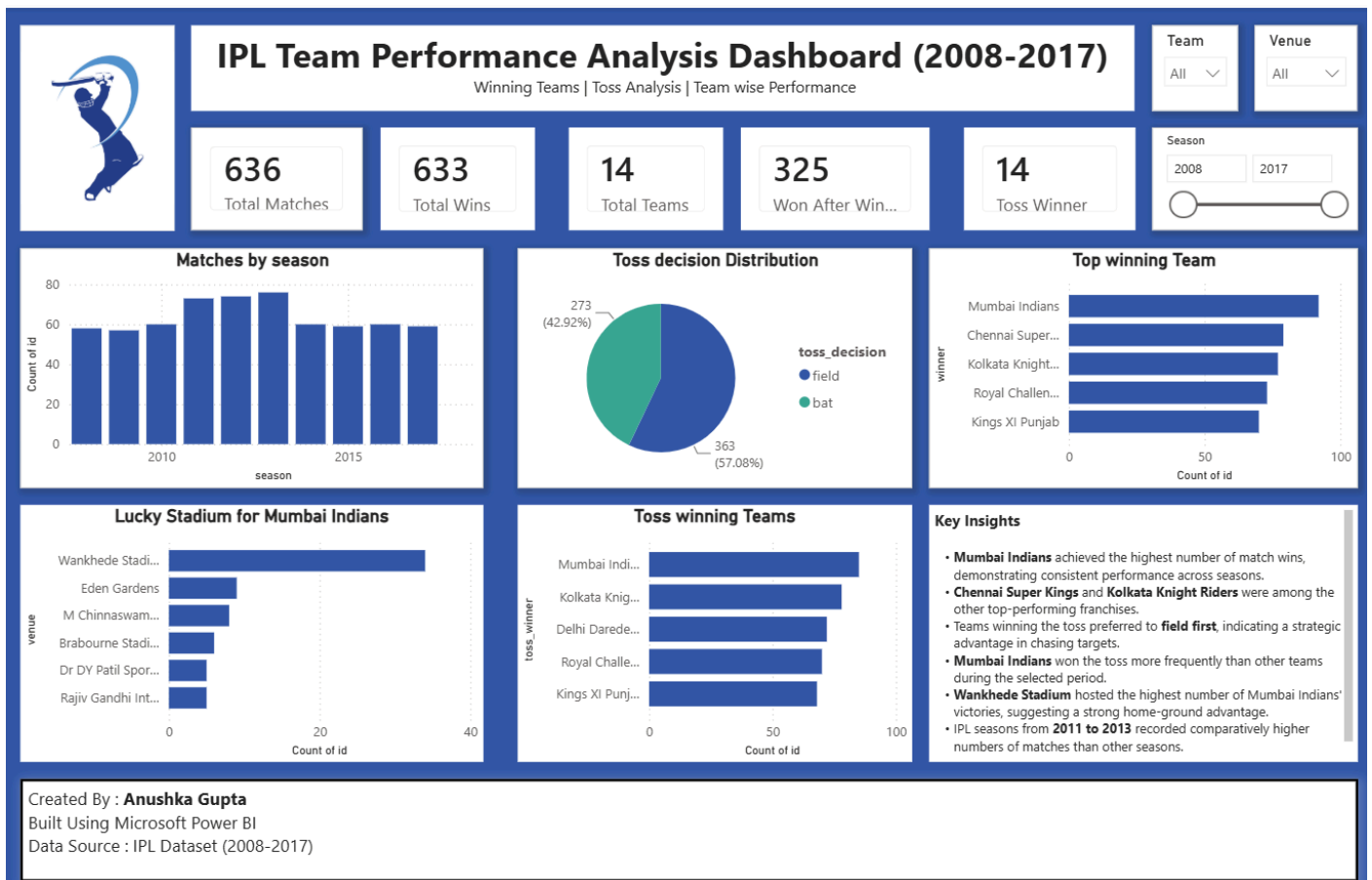


Figure 3: Team Performance Dashboard

Player Performance Dashboard Analysis

The **Player Performance Dashboard** presents a comprehensive analysis of individual player contributions across batting, bowling, and fielding during the IPL seasons from **2008 to 2017**. It highlights the tournament's top performers and enables easy comparison of key player statistics.

Batting Performance

The dashboard indicates that **Suresh Raina** scored the highest number of runs among the players displayed, reflecting his consistency and reliability throughout the tournament. **Virat Kohli**, **Rohit Sharma**, **Gautam Gambhir**, and **Robin Uthappa** also feature among the leading run scorers, demonstrating their significant contributions to their respective teams.

The boundary analysis further shows that **Gautam Gambhir** hit the highest number of fours, while **Chris Gayle** dominated in terms of sixes, highlighting his aggressive batting style and ability to score quickly.

Bowling Performance

Among the bowlers, **Lasith Malinga** recorded the highest number of wickets, making him one of the most successful bowlers during the selected period. Other leading wicket-takers also played an important role in restricting opposition teams and influencing match outcomes.

Fielding Performance

The fielding statistics reveal that **Dinesh Karthik** recorded the highest number of catches and dismissals among the featured players. This demonstrates the importance of fielding in modern T20 cricket, where every run saved and every catch taken can significantly impact the result of a match.

Overall, Player Contribution

The dashboard shows that consistent performers across multiple seasons had the greatest impact on the IPL. Players who regularly contributed with the bat, ball, or in

the field played a vital role in their team's overall success. The analysis also highlights the importance of experienced players who maintained high performance levels throughout the tournament.

Key Insights

- **Suresh Raina** emerged as the leading run scorer among the players displayed in the dashboard.
- **Lasith Malinga** recorded the highest number of wickets, demonstrating exceptional bowling consistency.
- **Chris Gayle** led the six-hitting statistics, reflecting his explosive batting approach.
- **Gautam Gambhir** scored the highest number of fours, indicating his ability to build innings through consistent stroke play.
- **Dinesh Karthik** made the highest fielding contribution, emphasizing the value of strong wicketkeeping and fielding skills.
- Experienced players consistently delivered strong performances across multiple IPL seasons.

Recommendations

Based on the player performance analysis, the following recommendations are suggested:

- Franchise teams should prioritize retaining **consistent performers** who contribute over multiple seasons.
- Batting strategies should balance **aggressive hitters** with **reliable run scorers** to strengthen the batting lineup.
- Teams should continue investing in **specialist bowlers**, as wicket-taking ability has a major influence on match outcomes.
- Greater emphasis should be placed on **fielding and wicketkeeping**, as these can significantly affect the result of close matches.
- Regular performance analysis using dashboards can help identify player strengths, monitor form, and support informed team selection.

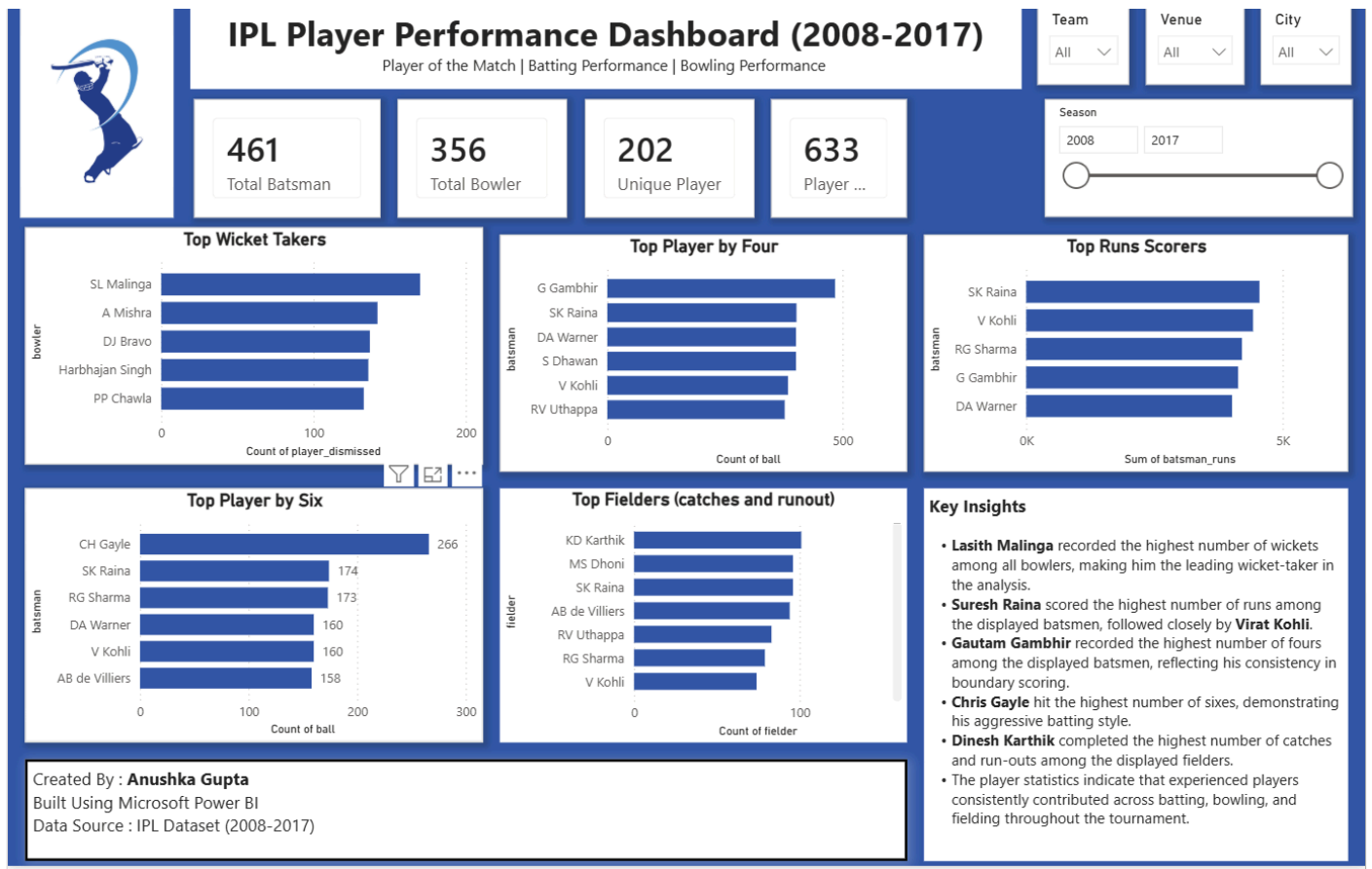


Figure 4: Player Performance Dashboard

Executive Dashboard Description

The Executive Dashboard serves as the main summary page of the IPL Analysis Dashboard. It provides a high-level overview of tournament performance by combining key performance indicators (KPIs), season-wise trends, team performance, player statistics, venue analysis, and interactive filters into a single dashboard. The dashboard is designed to help users quickly understand important insights before exploring detailed Team Performance and Player Performance dashboards.

The dashboard contains KPI cards displaying Total Matches (636), Total Players (202), Highest Team Score (263), Lowest Team Score (2), and Total Teams (14). It also includes visualizations showing season-wise match distribution, top run scorers, top winning teams, top host cities, and top wicket takers. Interactive slicers for Team, Venue, City, and Season allow users to perform customized analysis based on their requirements.

Key Executive Insights

- Mumbai Indians recorded the highest number of match victories during the selected period.
- Suresh Raina emerged as the leading run scorer, followed by Virat Kohli and Rohit Sharma.
- Zaheer Khan recorded the highest number of wickets among the displayed bowlers.
- Mumbai hosted the highest number of IPL matches.
- The highest number of matches was played during the 2013 season.
- Interactive filters enable dynamic comparison across teams, venues, cities, and seasons.

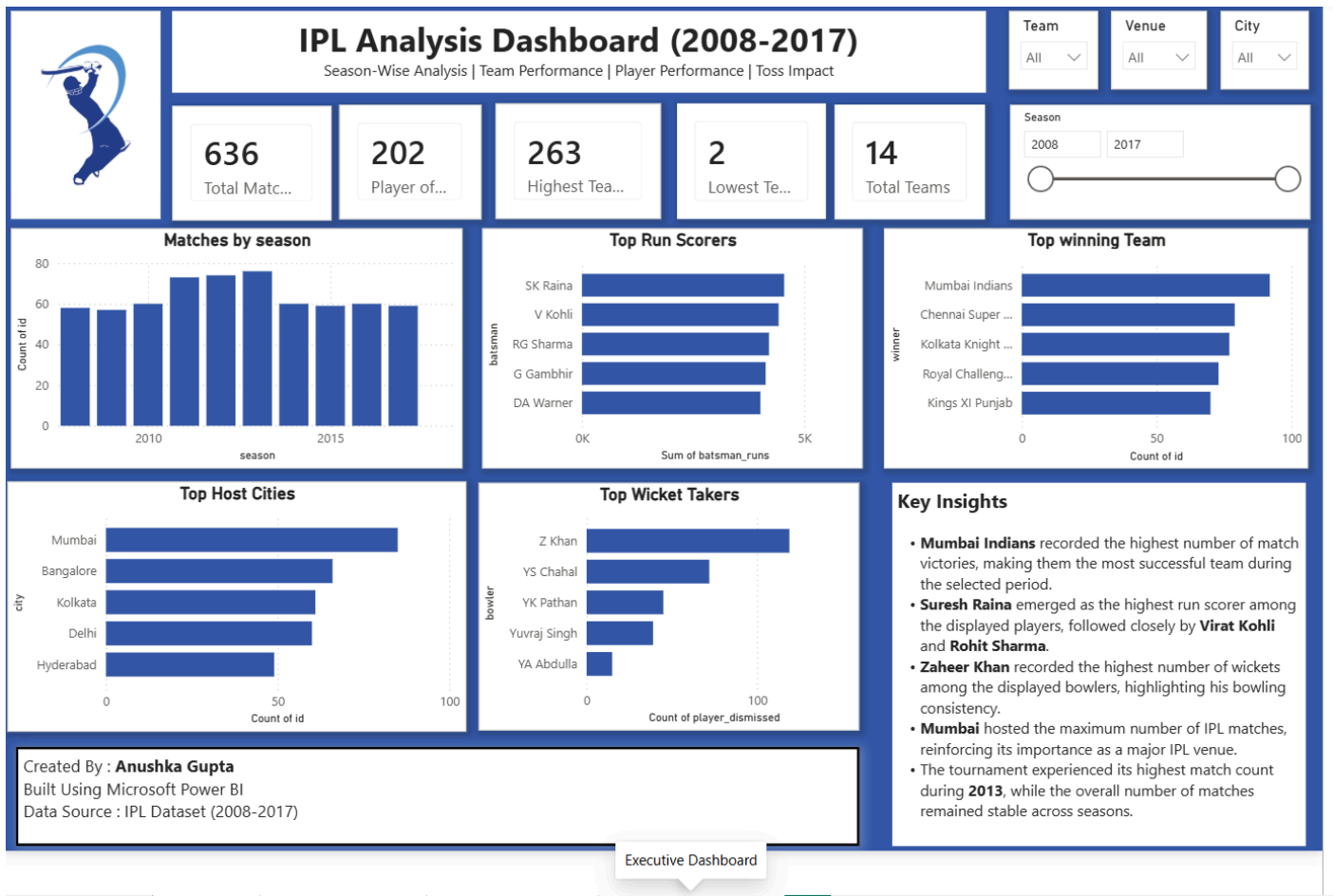


FIGURE 5: IPL ANALYSIS DASHBOARD

OVERALL FINDINGS & RECOMMENDATIONS

Overall Findings

The IPL Dashboard successfully transforms raw match data into meaningful visual insights, enabling a clear understanding of tournament trends, team performance, and player achievements from **2008 to 2017**. By integrating interactive visualizations with key performance indicators (KPIs), the dashboard provides an efficient platform for analysing historical IPL data.

The analysis reveals that **Mumbai Indians** were the most successful franchise during the selected period, consistently recording the highest number of match victories. The team also performed strongly in toss wins and home-ground matches, highlighting the importance of consistency and venue familiarity in achieving long-term success.

From the player analysis, **Suresh Raina** emerged as the leading run scorer, while **Lasith Malinga** dominated the bowling statistics with the highest number of wickets. **Chris Gayle** stood out as the most powerful boundary hitter, whereas **Dinesh Karthik** made a significant contribution through his fielding performances. These findings demonstrate that success in T20 cricket depends on balanced contributions from batting, bowling, and fielding units.

The tournament analysis also indicates that teams preferred **fielding first after winning the toss**, reflecting a strategic inclination toward chasing targets. Additionally, almost all matches concluded with normal results, emphasizing the consistency and competitiveness of the IPL throughout the selected seasons. Mumbai emerged as the leading host city, reinforcing its importance as a major IPL venue.

Overall Recommendations

Based on the analysis, the following recommendations are suggested:

- Use historical performance data to improve **match planning and opponent analysis**.
- Develop **venue-specific strategies**, as team performance often varies across different stadiums.
- Combine **toss decisions** with pitch reports and weather conditions to improve match strategy.
- Retain players who demonstrate **consistent performance over multiple seasons**, rather than relying only on short-term form.
- Give equal importance to **batting, bowling, and fielding**, as balanced team performance leads to better results.
- Regularly update the dashboard with new IPL seasons to monitor changing trends and player performance.
- Utilize interactive dashboards to support **data-driven decision-making** for coaches, analysts, and franchise management.
- Continue leveraging Business Intelligence tools such as **Microsoft Power BI** to simplify complex datasets and improve performance evaluation.

Conclusion

The **IPL Analysis Dashboard (2008–2017)** successfully demonstrates how **Microsoft Power BI** can be used to convert large volumes of cricket data into meaningful and interactive visual reports. By analysing ten seasons of IPL data, the dashboard provides valuable insights into tournament trends, team performance, player achievements, toss decisions, and venue statistics.

The analysis identified **Mumbai Indians** as the most successful team during the selected period, while **Suresh Raina**, **Lasith Malinga**, and **Dinesh Karthik** emerged as leading performers in batting, bowling, and fielding respectively. The findings also revealed that teams generally preferred **fielding first after winning the toss**, and that the majority of matches concluded with normal results. Venue analysis further highlighted **Mumbai** as one of the most prominent host cities throughout the tournament.

This project demonstrates the importance of **Business Intelligence (BI)** and **data visualization** in simplifying complex datasets and presenting information in a clear, interactive, and decision-oriented manner. The dashboard allows users to identify performance trends quickly, compare statistics efficiently, and support strategic planning using historical data.

Overall, the project achieved its objective of transforming IPL match data into a comprehensive analytical dashboard. It showcases the practical application of Power BI in sports analytics and highlights how data-driven insights can assist coaches, franchise management, analysts, tournament organizers, and cricket enthusiasts in making informed decisions.